**MQM — Multi-Quantum Mode Transmission**

**"Big ALU Quantum Jump: MQM-Based Ultra-Fast HBM Transfer with Zero-Heat Architecture"**

**Transitioning CPU/GPU/NPU architectures from single-bit line processing to multi-dimensional MQM-based data transmission and processing.**

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## 💡 Core Technology: 3D-Mapping Transfer

A revolutionary data transmission method that maps information into a **3D Symbol Space** within a single cycle.

**A single data symbol = ( Frequency, Amplitude, Phase )**

1-cycle data → 3D-mapping → High-speed transfer

$$S(t) = A \cdot \sin(2\pi f t + \phi)$$

| Dimension | Role |
|---|---|
| **Frequency (f)** | Multi-carrier modulation (OFDM) |
| **Amplitude (A)** | Multi-level signaling (PAM/MQM) |
| **Phase (φ)** | Phase shift keying |

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## 📐 Core Formula

Data Transfer Throughput = [ Frequency (n bits) + Amplitude (m bits) ] × 2^p

| Symbol | Description | Example |
|---|---|---|
| n | Bits encoded in the **Frequency** dimension | 8 bits → 256 states |
| m | Bits encoded in the **Amplitude** dimension | 8 bits → 256 levels |
| p | **Phase bits** | 3 bits |
| 2^p | **Phase states** (= Phase Number) | 3 bits → **8 states** |

**Key Point:** Phase is defined by **bit count (p)**, yielding **2^p independent phase slots**. With p=3: 3 phase bits → **8 phase states (0°, 45°, 90°, 135°, 180°, 225°, 270°, 315°)** → **(n + m) × 2^p bits transmitted per cycle, per channel.**

### 📊 Phase bits → Phase states

| Phase bits (p) | Phase states (2^p) |
|---|---|
| 1 | 2 |
| 2 | 4 |
| **3** | **8** |
| 4 | 16 |
| 8 | 256 |

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### 📊 Efficiency Example (n=8, m=8, p=3)

| Dimension | Bits | States |
|---|---|---|
| **Phase (φ)** | 3 bits | 8 independent slots (0° ~ 315°) |
| **Frequency (f)** | 8 bits | 256 states per slot |
| **Amplitude (A)** | 8 bits | 256 levels per slot |

**Total capacity: (8 + 8) × 2^3 = 16 × 8 = 128 bits (16 Bytes) per cycle**

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### 🔧 Designer-Defined Bit Sequence Mapping

The assignment of n bits and m bits to the **leading** or **trailing** bit sequence is **freely chosen by the system designer**:

| Option | Leading bits | Trailing bits |
|---|---|---|
| **Option A** | → Frequency (n bits) | → Amplitude (m bits) |
| **Option B** | → Amplitude (m bits) | → Frequency (n bits) |

**Key Point:** MQM does **not** mandate a fixed mapping — the designer selects freely based on hardware constraints and optimization goals.

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## 📜 MQM Design Freedom — 6 Configuration Options

In MQM, **Frequency (F), Amplitude (A), Phase (P)** can each serve as either **Data** or **State**. The system designer chooses freely based on hardware environment.

**Case 1: State × 1 + Data × 2 (3 options)**

| # State (배율) | Data 1 | Data 2 | Formula |
|---|---|---|---|
| **1** Phase(z bit) | Frequency | Amplitude | [F(n) + A(m)] × 2^z ← **기본 MQM** |
| **2** Frequency(n bits) | Phase | Amplitude | [Phase(z) + A(m)] × 2^n |
| **3** Amplitude(m bits ) | Frequency | Phase | [F(n) + Phase(z)] × 2^m |

**→ ex) Data 1, Data 2 : 1st byte (Data 2) 2nd byte(Data 1) 등 적절히 바꿔서 설계가능**

**→ ex) Data 1, Data 2 : 1st byte (Data 1) 2nd byte (Data 2) 등 적절히 바꿔서 설계가능**

**Case 2: State × 2 + Data × 1 (3 options)**

| # State 1 | State 2 | Data | Formula |
|---|---|---|---|
| **4** Phase(z bits) | Frequency(n bits) | Amplitude( m bits) | A(m) × 2^z × 2^n |
| **5** Phase(z bits) | Amplitude( m bits) | Frequency(n bits) | F(n) × 2^z × 2^m |
| **6** Frequency(n bits) | Amplitude( m bits) | Phase(z bits) | P(z) × 2^n × 2^m |

**Case 3: Data × 3 (6 options)**

| # Data 1 | Data 2 | Data 3 | Formula |
|---|---|---|---|
| 1 Phase(z bits) | Frequency(n bits) | Amplitude( m bits) | 2^z × 2^n x 2^m |
| 2 Phase(z bits) | Amplitude( m bits) | Frequency(n bits) | 2^z × 2^m x 2^n |
| 3 Frequency(n bits) | Amplitude( m bits) | Phase(z bits) | 2^n × 2^m x 2^z |
| 4 Frequency(n bits) | Phase(z bits) | Amplitude( m bits) | 2^n × 2^z x 2^m |
| 5 Amplitude( m bits) | Frequency(n bits) | Phase(z bits) | 2^m × 2^n x 2^z |
| **6** Amplitude( m bits) | Phase(z bits) | Frequency(n bits) | 2^m × 2^z x 2^n |

**Key Point:** There is no fixed rule — the designer selects the optimal configuration based on hardware constraints (GHz/THz environment, heat budget, bandwidth target).

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### ⚡ Why MQM Reduces Heat

| Feature | Benefit |
|---|---|
| Multi-dimensional encoding | More bits per cycle → **fewer switching events** |
| Fewer switching events | → **Less heat generated** |
| Lower heat | → Ideal for **HBM, CPU/GPU internal buses** |

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### 📈 Channel Scaling: Quantum Jump Extension

| Channels | Register Size | Throughput gain (vs 64-bit CPU) |
|---|---|---|
| 64 ch | 32 KB | 4,096× |
| 128 ch | 64 KB | 8,192× |
| 256 ch | 128 KB | 16,384× |
| 1,024 ch | 512 KB | 65,536× |
| 4,096 ch | 2 MB | 262,144× |

4,096 channels (n=8, m=8, p=8) achieves data capacity comparable to a **21-qubit quantum computer**.

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### 📜 Implementation Pipeline

1. **Compression** — Raw data → Huffman Encoding
2. **Encoding** — MQM Mapping Table → Fourier Encoding (Superposition)
3. **Transmission** — Complex waveform over HBM/Data Channel
4. **Decoding** — Fourier Decoding → MQM Reverse Mapping
5. **Restoration** — Huffman Decoding → Raw Data for CPU/GPU

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📜 **IP & License Notice: This technology is released as prior art. The author waives all patent rights.**

- Free open-source — no patent restrictions
- Anyone may implement, use, or build upon MQM freely
- Any third-party attempt to patent this logic is strongly opposed

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## 👤 Author & Credits


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- Community: https://cafe.daum.net/scwpark


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## ☕ Buy Me a Coffee

MQM is free and open-source forever — no patent, no fee. If MQM helped you, a cup of coffee keeps this old doctor-inventor going! 🦋

👉 Buy Me a Coffee

**Or directly via Shinhan Bank (신한은행):**

- 예금주: PARK SUNG TAE (박성태)
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"I had this dream in the 1980s. Forty years later, I'm giving it to the world for free. A coffee would be nice though." ☕ 😄